Project Initialization and Planning Phase

Date	02 July 2026
Team ID	xxxxxxx
Project Title	**OptiCrop: Smart Agricultural Production Optimization Engine**
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report :

The proposed Smart Agricultural Production Optimization System (OptiCrop) aims to assist farmers in selecting the most suitable crop using Machine Learning techniques. The system improves agricultural decision making by analyzing soil and environmental parameters and predicting the best crop in real time. This helps farmers increase yield, reduce uncertainty, and adopt datadriven farming practices.

Project Overview	
Objective	The primary objective of this project is to develop an intelligent system that predicts the most suitable crop for cultivation based on soil nutrients (N, P, K), temperature, humidity, pH level, and rainfall using Machine Learning algorithms.
Scope	The scope of this project includes data collection, preprocessing, model training, evaluation, and deployment of a Flask based web application that provides real time crop prediction and recommendations for farmers.
Problem Statement	
Description	Traditional crop selection methods rely on farmer experience and manual judgment, which may lead to low productivity and incorrect crop choices. This project addresses these limitations by introducing an automated ML based crop recommendation system.
Impact	Implementing this system improves agricultural productivity, reduces crop failure risk, optimizes resource usage, and supports data driven decision making in farming.
Proposed Solution	
Approach	The system uses Machine Learning algorithms to analyze agricultural data such as soil nutrients and weather conditions to predict the most suitable crop. The trained model is integrated into a Flask web application for easy and interactive user access.

Key Features	• Machine learning based crop prediction system • Real time input and instant crop recommendation • Userfriendly Flask web interface • High accuracy prediction using trained models • Reduces dependency on manual farming decisions

Resource Requirements:

Resource Type	***Description***	***Specification/Allocation***
Hardware		
Computing Resources	Processing power required for model training and execution	CPU / GPU (T4 GPU for faster training)
Memory	RAM specifications	8 GB RAM
Storage	Disk space for dataset, model files, and logs	1 TB SSD
Software		
Frameworks	Backend development framework	Flask
Libraries	Python libraries for ML and data processing	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source of training and testing data	Crop Recommendation Dataset (Kaggle)